

Chapter 12

Gaining Support Within Companies for Collaboration



Chris Ramming

12.1 Introduction and Terminology

In general, for companies that engage with universities, some employees can be considered “**practitioners**” of university engagement while others can be considered “**stakeholders**” in university engagement. Practitioners need to justify and implement a material investment comprising direct financial support, in-kind contributions (such as equipment and facilities), and/or employee time. Stakeholders are the beneficiaries of the practitioners’ work and typically need to participate in some way to receive that benefit (e.g., with personnel and resources even if on an informal or intermittent basis). We will use the term “**collaboration**” to describe the activities of engagement, for example, working together on a specific research project, hosting an intern, developing curricula, or advising students. Typically, practitioners have a deep understanding of both the university and the company, and they use that knowledge to motivate and orchestrate collaborations that advance the goals of stakeholders on both sides. This chapter is aimed primarily at supporting corporate practitioners of university collaboration, and indirectly at helping university-side collaborators understand the context and purpose of their collaboration from the broader corporate perspective.

Viewed through an organizational lens, practitioners are often members of a broader “**Academic Engagement Team**” (AET) with budgeted responsibility for leading and managing university collaboration in some form (see Callout 1). Some large companies have multiple de facto AETs focused on different facets of collaboration, for example, research, talent acquisition, workforce development, strategic marketing, and corporate social responsibility. Some very large companies also have an umbrella “university relations” AET that helps coordinate the more

C. Ramming (✉)
Portola Valley, CA, USA

specialized AETs and that can serve as a one-stop interface to universities engaged in multiple ways with a given company.

In some cases, the actual corporate collaborators are formally part of an internal AET, but in many cases, those collaborators are part of a stakeholder organization (such as a specific product team or business unit) that is intended to benefit from the collaboration activities. For example, a business unit leader may be a stakeholder in research that could potentially influence product roadmaps, and given an opportunity for potential competitive advantage may be willing to formally dedicate and support activities needed to achieve the benefits of university R&D collaboration. In some cases, an AET practitioner will work to build stakeholder support in a “top down” manner (starting with the executive stakeholders), in other cases the support is built “bottom up” (through engineering stakeholders, product manager stakeholders, and so on).

A theory of effective university–industry (UI) collaboration would be straightforward to describe but for a dizzying array of variations on the theme stemming from

- Corporate strategy (manifested in the balance of emphasis on different forms of collaboration)
- Organizational design (especially degree to which university collaboration is managed by a centralized organization)
- The maturity and role of the company in its industry (which can influence the scope of university collaboration initiatives)
- The manner and ways in which other collaborators (government, nonprofits, and other companies) are incorporated into the collaboration
- the fundamental purposes and cultures of academia and industry (often leading to expectation misalignment that needs to be anticipated in the collaboration design)

After introducing certain basic facets of university collaboration, this chapter will discuss complicating factors and their implications for the design, operation, and justification of AETs. This chapter will also highlight a case study that illustrates the application of suggested considerations.

12.2 Opportunities and Challenges

To make an effective case for university collaboration, the practitioner of UI collaboration needs to be an organizational “polymath”: technically astute, alert to corporate philosophy, capable of navigating a variety of organizational designs, adept at building support of key stakeholders, and strategically aware of the company’s maturity and its role in a broader industry context. Before diving into practical solutions and approaches, it is worthwhile to briefly tour some of the issues that a practitioner needs to consider.

12.2.1 Alignment with Corporate Philosophy

To drive an effective AET, the corporate practitioner needs to be sensitive to the broader philosophy of their enterprise. Industry in the United States is primarily organized around the profit motive, and therefore, the returns on investment (ROI) of university collaboration need to be rationalized in terms of financial or strategic motivations, competitive advantage, and the opportunity costs of alternative investments. However, individual corporate philosophy also plays a role, and in some cases, the broader corporate philosophy can be either supportive or antithetical to specific forms of university engagement. The deciding factor in the strength or absence of university-facing functions is often the CEO's approach to business. Here are several illustrations of the relationship between corporate philosophy and university engagement.

R&D Whether and how a company supports disruptive research depends in part on the company's philosophy of growth. Does it deliberately fund "organic innovation," that is, home-grown new products and services? Or does it grow primarily through "inorganic innovation," that is, the acquisition of companies with established products and services? University collaborations can be relevant in both settings, but the focus must be aligned with the broader corporate philosophy. For example, to focus on organic growth the practitioner might prioritize "research translation" metrics and incubation-related activities. To focus on inorganic growth, the practitioner might focus on scouting entrepreneurial faculty and on internal education to create a more informed corporate "buyer" of acquisition targets early in their lifecycle.

Talent Acquisition In some cases, the hiring returns to a "talent acquisition" organization may be easy to quantify and justify. For example, a company establishing a new regional engineering office near a potential "talent feeder" university may see an attractive recruiting ROI from engagement with that university. Similarly, even without regional considerations, an organization may find that it needs to grow in skills that are best cultivated in a university setting. But executive preference can also be a deciding factor in the decision not to collaborate: for example, some companies deliberately hire only experienced employees and do not recruit new graduates. Clearly, the decision to engage in technical collaborations and/or to fund university programs for a talent acquisition purpose is entirely dependent on broader corporate hiring policies and philosophy.

Corporate Social Responsibility In some cases, a company may be operated on tenets of "Stakeholder Capitalism" that seeks to benefit employees and society as well as shareholders. Such companies may be more willing to engage in long-term value creation even at the expense of short-term profits. On the other extreme, some companies adhere more to Milton Friedman's view that "The Social Responsibility of a Business is to Increase its Profits," which in some interpretations could put a skeptical eye on investments in certain kinds of university initiatives (e.g., diversity

and inclusion initiatives, speculative longer-term research), or perhaps simply favor forms of university engagement with more obvious marketing revenue drivers such as product placement and professional curriculum development.

The bottom line is that practitioners need to clearly understand the prevailing corporate philosophy as they seek support for AET initiatives.

12.2.2 Metrics: Key Performance and Effectiveness Indicators

To be effective in corporate settings, the practitioner needs to develop, monitor, and produce results in terms of metrics aligned to corporate priorities. A typical challenge for UI engagement is therefore to identify metrics and performance indicators. These indicators serve two purposes: first, to help justify the investment in university collaborations, and second, to serve as management tools. However, picking metrics is notoriously challenging for university engagement:

- The benefits of university engagement are difficult to reduce to financial metrics such as expected net present value and revenue. Without such metrics, the practitioner faces a thorny challenge in enlisting the support of business unit leaders and arguing for a budget to invest in collaborations.
- The total value of an engagement is the sum of incommensurable constituent metrics corresponding to the different facets of engagement (hiring, research outputs, strategic marketing, workforce development, employee development, etc.). As a result, it can be challenging to make an investment case to mid-level managers whose scope extends only to a subset of such facets. Correspondingly, the executive whose scope extends across these facets tends to be high in the organization, and may even be the CEO. This points back to the importance of aligning with corporate philosophy in the design of an AET.
- The business impact of a given investment can take many years to realize. This lag induces a reliance on leading indicators that by themselves may not be persuasive to business unit stakeholders (e.g., research publications may be a leading indicator of competitive differentiation, but do not always result in business impact).

In addition to these conceptual challenges, there are many implementation challenges in tracking university collaboration metrics. For example, at the infrastructure level, corporate HR systems may not be configured to capture information about whether a new hire was connected to a prior research collaboration.

More generally, the results of university collaboration (especially technical results) wend a long path from research groups to the benefiting product groups. Tracking the provenance of collaborative contributions is particularly tricky as ideas mature and as the custodian of those ideas move from group to group. Even when an AET initiates a university collaboration, other groups are ultimately responsible

for realizing business value, and the provenance of collaborative results can be forgotten along the way.

The bottom line is that an AET will be more successful if its plans and impact can be appreciated through the lens of relevant objective metrics.

12.2.3 Funding: Allocating and Managing Budgets

The corporate practitioner can “follow the money” to understand and support the company’s goals. In general, corporate budgets are carefully allocated to reflect business priorities. Clearly the decision to allocate funding in a particular way and to a particular organization or suborganization is an important decision. For those joining a new organization and trying to understand its philosophy of university engagement, the funding landscape can be analyzed to quickly appreciate an organization’s priorities and perspective on academic engagement. For example, funding through a corporate foundation or donor-advised fund clearly has a philanthropic, social responsibility component. In contrast, funding through a business unit or a functional group (HR, marketing) tends to be very focused on extending the goals of that business or functional group. And funding through a centralized organization such as the CTO office will suggest interest in looking beyond current lines of business.

Importantly, for those working to evolve a company’s approach to university engagement, reallocating the budget (however contentious it may become) can be a key management tool. Moving budget from one organization to another is one way to re-prioritize engagements to align with the purpose of the receiving organization.

The bottom line is that in the long run the returns on university collaboration need to track to the purposes for which funding has been allocated, or that funding can be at risk. The practitioner can sometimes use budget allocations to align effort.

12.2.4 Governance: Deciding How to Invest

The practitioner can extend support for university engagement by creating governance structures that extend responsibilities and ownership to a broader set of stakeholders and supporters. In most corporate settings there is a close correspondence between budgets and decision authority over that budget. However, for centralized budgets in particular, and for university engagement programs that are not tied to quarterly profit and loss constraints, practitioners often find benefit in establishing governance structures that incorporate stakeholder input in the investment decisions.

For example, a centralized research AET may wish to empower a “technology council” or an “advisory council” or a “board of advisors” that reviews and recommends investment proposals, strategic plans, and research collaboration progress.

Although such councils are inherently inefficient in the sense that they complicate decision-making and can be viewed as mere bureaucracy, councils comprising influential executives and technical leaders can nonetheless be valuable allies for building awareness and support for collaborations and individual project results. Ideally, council participants will both evangelize the program and help marshal the resources of their organizations around activities such as research translation. Deciding how to organize such governance structures is a key opportunity and sometimes a challenge. Once established, changing the governance approach can be as important a tool for evolution as reallocating the underlying budget.

The bottom line is that sharing responsibility and authority can increase buy-in and impact in university collaborations.

12.2.5 Organizational Design: Optimizing the AET Overlay

The practitioner will be more successful when the company's AET(s) are thoughtfully incorporated into its organizational design. However, the multifaceted range of corporate interests (hiring, technology development, CSR, etc.) focused on a single category of partner (universities) means that there are many possible ways to design a company's university-facing functions. The practitioner needs to consider how a given design affects their approach to university collaboration.

AETs are often implemented as "overlays" on other corporate business units (see Callout 1). Fully realizing the benefits of university collaboration therefore often requires practitioners to work effectively in a matrixed, highly decentralized organization. Importantly, the position of a practitioner within the larger company structure can be a factor in their effectiveness. "Centralized" practitioners may be more distant from day-to-day business priorities and less well positioned to identify and advocate for the most aligned university collaborations. Conversely, decentralized practitioners may be less aware and less fully incentivized or chartered to identify and advocate for university collaborations that serve the company but fall outside of an individual business unit purview.

Bottom line: practitioners need to consider and compensate for the strengths and weaknesses of their particular corporate design.

12.2.6 Evolution: Right Opportunities at the Right Time in a Company's Lifecycle

The practitioner needs to be aware of the broader context of their company in order to pursue the most impactful strategic opportunities. An often-overlooked aspect of university-industry engagement is that companies evolve over time, sometimes with sharp inflection points in the university relationship. By thinking broadly about the evolving balance of corporate needs and the consequences for university

collaboration, it may be possible to anticipate and productively navigate change. For example:

- Startups face pressures that are different from those of an established company.
- Companies in a mature industry may have research priorities that differ from those in an emerging technology area.
- Conglomerates may have broader research interests and a different organizational structure than a company that is focused on a single line of business.
- Market leaders in tech-heavy industry segments may invest differently than new entrants (e.g., in pre-competitive research vs proprietary differentiation); even the format of the research may differ in this context. For example, a market leader may have a lower incentive to support open-source solutions than those trying to “catch up.”

Model of a Corporation with Academic Engagement Teams

There are many ways to organize a company and many ways to organize an AET within a company. A generic model of company organization contains one or more largely independent “business units” (BUs, e.g., manufacturing, design, or software) and one or more centralized cross-BU functions under leaders such as the chief technology officer or human resources officer. Depending on the degree to which a company subscribes to Milton Friedman’s views on its desirable scope, there may also be a leader for corporate social responsibility to drive environmental sustainability, diversity, equity, inclusion, and other forms of responsibility. Some companies minimize centralized functions by design, so their individual business units handle those functions; others seek synergies across lines of business and leverage centralized organizations to drive synergies. An AET organization needs to be embedded in (aka overlaid on) the *de facto* corporate structure. It can be arranged in myriad ways. Some AET practitioners are “decentralized”, by which we mean embedded in their stakeholder business units. Other AET practitioners may be part of a centralized organization such as the CTO office (logical for research-focused practitioners) or the HR office (logical for hiring-focused practitioners). A practitioner’s university-related responsibilities may be full-time or part time, and may be formal or informal. A company’s practitioners may report to a single “centralized” leader or to separate leaders in their host organization. It is common for a company pendulum to swing between centralized and decentralized approaches as it evolves, reflecting the challenges of finding a balance between business-unit-specific needs and generalized, less tailored services. However, an AET is designed, the university-facing practitioner seeks to inspire and leverage collaboration from engineers, hiring managers, and others whose primary function and expertise does not involve university collaboration.

In this light, the successful practitioner should prepare to lead the organization's university-facing functions through growth and change.

Bottom line: it is essential to fully understand the company's industry context and maturity in order to make an effective case for the "right" university collaborations.

12.2.7 Timescale Harmonization: Corporate with University

The practitioner needs to be aware of structural and cultural differences between industry and universities in order to motivate successful engagements. One of the most fundamental issues in UI collaboration is the difference in timescales and the pace of progress. Industry partners often work on quarterly or annual planning cycles, whereas PhD students need roughly 3 years to complete their thesis work. Industry partners are focused on bringing ideas to market quickly and predictably, whereas not every university project yields commercializable results. Even mature research at a university can take years to bring to market.

Timescales for patent commercialization can vary widely, but as explained in a report (Pressman et al., 2022) prepared for the Biotechnology Innovation Organization (BIO) and the Association of University Technology Managers (AUTM), "there is considerable evidence that nonprofit technology is developed by its licensees for years after it is licensed but before products are sold." The report cites studies showing between 3 and 8 years are needed to commercialize a licensed invention, which patent itself has presumably taken several years to issue.

Evidently, in life sciences, innovation timelines can be even longer as it can take over 10 years to go from "Phase 1" to regulatory approval (Thomas et al., 2021). In other disciplines, such as information technology (cloud computing, networking, etc.) and open-source software (OSS) may accelerate innovation and serve as a coordinating artifact, despite differing timescales. Practitioners need to manage expectations accordingly in the context of their company's philosophy.

Bottom line: unless timescales are harmonized and expectations are set correctly on both sides, the practitioner will be challenged to motivate collaboration and deliver on realistic expectations.

12.2.8 Innovation: Bringing Research Ideas to Market

In many cases, the practitioner needs to have a systematic approach to "research translation" in order to motivate and operate collaborations with product impact. UI collaborations face a special challenge in managing innovation. Bringing a research concept (an "invention") to market ("innovation") requires significant corporate investment, often over the course of years. The responsibility for bringing a university-inspired idea to market often shifts to people outside of the university

collaboration apparatus (the AET). Setting the stage for a successful innovation trajectory is challenging, as is tracking the degree of impact on a new product once responsibility has been transferred to a product team. The timeframe for research translation can be years, and the evidence of impact can be diffused or lost as the fruits of a research collaboration evolve.

Bottom line: to maintain support and show evidence of impact, the practitioner needs to develop, retain, and articulate evidence of that impact over time.

12.2.9 Partnering: Recognizing When to Engage Government, Other Industry, and Nonprofits

The practitioner can often leverage and magnify funding and impact through broader external partnering, which can add to corporate appetite for university collaboration. Industry–university collaboration is often influenced or supported by governments, yielding what are sometimes called “triple helix” collaborations that come in the form of joint investment, joint solicitations, and a range of contractual approaches from contracting to cooperative agreements. Motivating these forms of collaboration requires an understanding of the benefits and tradeoffs relative to pure bilateral UI collaboration. Today, many governments are reflecting on how to maximize their nation’s investment in education and research (United States, 2022; National Science Foundation, 2022); optimizing UI partnerships is a topic of ongoing interest (Khargonekar et al., 2021). Section 12.3.8 discusses a present-day resurgence of interest in public–private partnerships.

Bottom line: the practitioner can be more effective when considering and creating options for funding and collaboration beyond bilateral university agreements.

12.3 Options, Solutions, and Impact

Touring the above-mentioned challenges and opportunities can help a practitioner inspire their colleagues with carefully chosen data and rationales for university collaboration. Here, we focus primarily on research collaborations; similar issues and analyses apply to workforce development, corporate social responsibility, and other facets of engagement.

12.3.1 Alignment with Corporate Priorities

As described in section 12.2.1, practitioners of UI collaboration serve the interests of the corporation and need to accommodate that philosophy, whatever it is or may be. Moreover, in a subordinate role, proactively addressing change is crucial. Some of the actions a practitioner can take include

- **Create programs that demonstrate returns in objective, quantifiable terms that reflect the corporation's priorities.** In general, it behooves the practitioner to design and maintain objective metrics of impact that substantiate the benefits of the engagement.
- **Anticipate changes of control and/or philosophy.** Especially for corporations that lean toward stakeholder capitalism, it can be helpful to anticipate the inevitable pendulum swings and forcing business cycles, which lead to company-wide scrutiny of its prioritized investments. By operating in a “worst case” mindset per Andy Grove’s famous motto, “Only the Paranoid Survive,” the UI practitioner can develop a living portfolio of results that can be used to educate new executives and respond effectively to the inevitable strategy reformulation “fire drills.”
- **Partner with business unit leaders/stakeholders.** In general, business unit leaders are acutely aware of profit-and-loss (P&L) drivers and, as such, are able to help gauge whether an AET and its programs contribute in a meaningful way. As implicit or explicit stakeholders in UI collaborations, their support can be crucial in justifying a central budget as well as augmenting that budget for BU-tailored initiatives.
- **Earn the support of internal technical community leaders.** To the extent a company’s strategy is based on leadership in the state-of-the-art, technical community leaders benefit disproportionately from insight into academic research frontiers. As such, the fellows, principal engineers, and those aspiring to become technical leaders see direct albeit sometimes intangible value in academic engagement. For example, when the corporate promotion philosophy includes technical leadership criteria, academic engagements can play an important part in helping those practitioners succeed. These largely intangible “human resource” returns can nonetheless contribute to support for centralized, longer-term UI investments.
- **Market UI programs internally.** In general, newsletters, workshops, and conferences serve the important function of raising awareness within a company and in turn creating new opportunities to leverage collaborations on behalf of business opportunities. Investing in internal marketing therefore is a crucial alignment technique.

Notwithstanding the importance of these “defensive” measures, it is also sometimes possible for practitioners/leaders of an AET to more proactively identify and shape corporate direction and philosophy. In particular, AET leaders have a seat at the discovery table and are privy to deep insight about the potential technical direction

of their industry. With a birds-eye view of that landscape, new business directions can sometimes be identified that afford the university research team a leadership position in shaping the company's future. Moreover, an experienced AET leader may be positioned to help a company understand when changes in its evolution can be supported with new AET activities (see Sect. 12.3.5). In this light, the chief technology officer (CTO) generally has sufficient position power to effect change and is often a key partner and stakeholder in translating related insights into corporate strategy effects.

12.3.2 Metrics

An important exercise for practitioners is to define relevant performance indicators as a way of aligning with the purpose and philosophy of the AET and its specific purpose within the company's priorities. These metrics would be defined at multiple levels: for the AET as a whole, for specific programs within the AET's purview, and for specific collaborations within a program. The metrics would unambiguously show the value of the research (measures of effectiveness) and serve as useful management tools along the road to business value (measures of performance). Ideally the metrics would be necessary and sufficient to characterize and manage the desired collaborations. In an ideal world, the raw metrics would be straightforward to implement without excessive overhead.

Because of lengthy research timescales, the first order of business (especially for a newly created AET) is to develop "leading indicators" of success that can be used to manage engagements. For example, academic publications and/or patents can be viewed as leading indicators of meaningful product impact given that the academic peer review process and the patent system have both been honed over decades to identify novel advances. A collaboration that is intended to produce non-incremental, disruptive innovation might further recognize only publications in top-tier conferences and journals, on the theory that the competition and quality bar for such conferences is higher. But even that higher bar does not assure impact, which can only be judged retroactively in part by "trailing indicators" such as citations by other researchers and actual use in revenue-generating products.

In the longer run, the important industry metric must be measured in financial terms and may not take place until long after the university engagement is complete, and it may not be easily recognizable as deriving from the collaboration. The practitioner therefore needs to find ways to track progress even after custody of a research concept has moved into a business unit. Special care should be taken to document successful research translation into revenue-generating products and services, as these become the quantifiable metric for justifying continued investment.

12.3.3 Funding Approaches

Common funding structures and their implications are listed in Table 12.1. These structures are not mutually exclusive and can reinforce each other. For example, companies that have both centralized funding (typically in a CTO office, as part of a corporate HR apparatus, etc.) and decentralized funding (in a business unit) may impose conditions on distributing that funding to universities. An example of such a condition might be a requirement for business units to provide matching funds as an indicator of commitment.

One responsibility and opportunity for AET practitioners, especially in a growing company, is to understand when specialized purposes have grown in importance such that a change to such funding structures is needed. Such changes may be

Table 12.1 Potential academic engagement team funding structures

Funding approach	Example pros	Example cons	Inferred AET values
Centralized funding via operating expense (OpEx) budget	• Less constrained by current line of business priorities/biases • Economies of scale in operating personnel	• Separation from business units, complicating research translation into product • Subject to short-term business cycle fluctuations	• Company wishes to pursue strategic objectives beyond quarter-to-quarter P&L
Centralized funding via a corporate foundation or donor-advised fund	• Less constrained by current line of business priorities/biases • Not subject to short-term OpEx budget fluctuations	• Cannot be used to accrue “prohibited benefits” such as Intellectual Property (IP) rights	• Company invests in corporate social responsibility
Distributed funding by business units via “operating expense” budget	• Line of business stakeholders improve odds of direct research translation (“innovation”) into products	• Not likely to lead to new lines of business	• Company expects concrete, short-term benefits from investment
Distributed or centralized funding by organizational functions (e.g., talent acquisition, R&D, CSR, marketing, training)	• Isolates facets of interest (e.g., hiring) and supports specialized engagements for that purpose	• Leads to AET fragmentation and complicated interfaces between different aspects of university collaboration	• Company values some particular facet of collaboration
Funding allocated to “research councils” representing specific technical disciplines contributing to the company’s success	• Isolates facets of technical interest (e.g., networking, storage, architecture) and builds internal alignment around related research	• Difficult to evolve budgets as new technical priorities emerge	• Company values a particular technology stack

induced by factors such as growth in the company or changes in the external environment and competitive landscape. The practitioner can then take a leadership role in articulating the need, defining the approach, and implementing the solution.

For example, it may be that curriculum, product donation, and talent aspects of university engagement are all initially handled by the same AET team that is focused on research engagements. However, as the program or company grows, it may prove more efficient to fund and operate those facets through the work of specialized teams. Alternatively, it may be that a company initially grows *de facto* AETs in several of its individual business units and has no centralized AET with independent funding. This arrangement may not work well if the company enters a phase where it is aggressively seeking growth through organic innovation: the business units will be predisposed to bolster their existing lines of business instead of seeking new opportunities.

Another motivation for centralization, as seen in the case study, can be the need to respond to a company-wide strategic challenge that would not be “in scope” for AETs organized by technical discipline.

12.3.4 Using Agreement Vehicles to Align UI Timescales

As a rule of thumb, collaborations with clearly defined technical goals and short time horizons are the easiest to justify, and that justification will typically be best described and evaluated by engineering team stakeholders in a company’s decentralized business units. The research topic itself can also dictate whether the engagement is best conducted in a decentralized or centralized business unit. Research related to improving existing products will most naturally be homed in an existing business unit. In contrast, those business units may be unable to support or even appreciate research that is disruptive or adjacent to their products and services. Agreements need to be tailored to the timescale of the engagement.

Consulting At one extreme end of the spectrum are consulting agreements (with faculty or students) meant to solve a specific product engineering problem. Often involving confidential parameters and well-defined objectives with clear timelines for impact, consulting agreements are best motivated and managed in a decentralized fashion. Engineering teams can take the lead on initiating and defining such collaborations and generally have the mechanisms and processes to onboard the faculty without involving the AET practitioner. Even so, a centralized AET team may be helpful in tracking and understanding the frequency, number, and participants in such consulting arrangements as they pertain to the overall value of a specific university relationship.

Interns and Visiting Faculty Student internships and visiting/sabbatical faculty arrangements are typically more open ended than consulting agreements, but still need to be hosted in research and distributed engineering teams to be impactful. The

technical value of these collaborations is often best assessed by those directly collaborating. There can be crossover motivation and value to (typically centralized) talent acquisition AETs, as intern programs often serve to feed an early-stage hiring pipeline. Importantly, interns focused on longer-term, more open-ended research collaborations are often on an academic career trajectory. It can be viewed as a “win” even when these interns become faculty instead of employees, especially when they continue to collaborate as faculty. The timescales for developing early-career faculty relationships span many years and require a long-term mindset on the part of the AET’s corporate stakeholders.

Joint Appointments Joint faculty appointments are an emerging form of collaboration that concurrently tackle defined short-term proprietary business problems (in the role’s employee persona) and broader, longer-term research opportunities (in the role’s faculty persona). There is typically grass-roots industry motivation for joint faculty, and evaluating the success of an employee relationship is handled by day-to-day business processes. However, because the university is intimately concerned with the success of the arrangement, a centralized team may add value in supporting the university relationship in defining and executing the opportunity, and representing the joint employment as an element of a broader engagement with the partner university (UIDP, 2024).

Sponsored Research Agreements (SRAs) move further down the spectrum toward longer time horizons and increasingly open-ended problem formulations. These can take the form of individual faculty support, affiliate program engagement (Stanford, 2024), large-scale research centers (Tennenhouse, 2004/2016), and other formats. Maintaining corporate support for SRAs can be challenging for a practitioner because universities operate on longer timelines than corporate product development. For example, PhD programs involve three or more years of research, compared to product groups in some industries that operate on quarterly planning cycles. Moreover, the connection between university research and revenue is rarely direct.

One approach to building support for sponsored research collaborations is to handle the earliest phases of university research projects with a centralized team until it matures enough to be actionable by a business unit, and then bring it to the business unit’s attention for a follow-on phase of advanced development. With this approach, the business unit need not be distracted by early explorations, but at the same time an opportunity to shape and guide the research may be lost. The dilemma for engagement management can sometimes be handled in a lightweight way by inviting business unit personnel to provide strategic context to the university teams and convey enough strategic perspective to inform research directions. Tracking the progression of early-stage research is arguably best done by a centralized team, potentially using indicators discussed below in the research translation section.

12.3.5 Company Maturity

Importantly, as companies progress in their lifecycle from startup to maturity, their broader needs for university research collaboration can evolve significantly. The practitioner may benefit from considering the company and industry context when building support for collaboration. At one end of the spectrum are startups, and at the other, are established industry leaders.

At the risk of overgeneralization, startups seek to minimize technical risk by the time of their founding and are focused more on engineering than the discovery of additional fundamental breakthroughs. Thus, the need for university collaboration may not be urgent and, if conducted at all, the research may be focused on other benefits, such as talent acquisition, use case discovery, and solving complementary problems. In fact, open-ended research collaborations may be viewed as a distraction in a company's early days. For example, despite being strong proponents of research and innovation, Intel's founders are said to have discouraged research projects outside of their core product for fear that it would inspire people who, like themselves, left a prior employer to pursue an idea that could not be realized in the existing business model. Whether or not this story about Intel's founders is apocryphal, "brain drain" is a risk.

At the opposite end of the spectrum from startups, mature industry leaders have the scale and perspective to take responsibility for the overall health of their industry. It can be helpful to a practitioner to recognize when this shift takes place within the company or when shifts in the industry technology landscape take place. Often those shifts signal the opportunity to launch new and impactful collaborations. (See the semiconductor industry case study).

12.3.6 Research Translation

Research translation (aka "innovation," the process of moving discoveries into products, services, and practice) is notoriously challenging. Universities and internal research groups are traditionally focused on expanding the sum of human knowledge with new ideas and discoveries ("invention"). The research process itself is lengthy, requiring years to build expertise and years to make meaningful new contributions, subject to some probabilistic combination of sweat equity and serendipity. Moreover, putting those inventions into practice ("innovation") is equally challenging. The process of incorporating an invention into a new product or service is lengthy, fraught with technical risk (can the engineering team even incorporate the result?), market risk (will product managers discover a compelling "product-market fit"?), and go-to-market risk (will the company's sales apparatus be successful?). In general, it can take years to bring a research idea to market.

One key strategy for building an R&D pipeline is to use a funnel model in which a large number of initial exploratory engagements is managed primarily by

members of the AET, then winnowed down through systematic criteria until the concept and its relevance are mature enough to make a case for direct business unit stakeholder engagement. At that point responsibility for the collaboration may shift from the AET to the business unit. Potential indicators of progress in a funnel are illustrated in Fig. 12.1—such milestones can be tracked by an AET to help maintain support for the overall program and help prioritize projects to actively drive to the next phase. Importantly some ideas are compelling enough to skip intermediate stages, for example, a business unit may take direct interest in a compelling university breakthrough based solely on the university’s work.

Importantly, research translation within a company is not always the principal rationale for collaboration. In some cases, the research is aimed at eliminating bottlenecks in other parts of the technology ecosystem, for example, the parallel programming bottleneck illustrated in the semiconductor industry case study.

The relationship of university collaboration to inorganic growth is also interesting to consider. In some fields, such as computer science, leading faculty and students are often highly entrepreneurial. Thus, engaging in university collaboration can provide a window into the evolution of an emerging industry. Berkeley’s tradition of 5-year research centers has resulted both in technical advances such as Apache Spark (Apache Software Foundation, [n.d.](#)) and Ray (Berkeley, [n.d.](#)), which in turn underpinned in influential “unicorn” startups (Databricks, [2013](#)) (Anyscale, [2019](#)) that have fueled important disruptive progress (including OpenAI’s ChatGPT, which uses Ray (McManus, [2023](#))). In some cases, faculty startups may prove to be competitors, and in some cases partners, and in some cases acquisition targets in their own right.

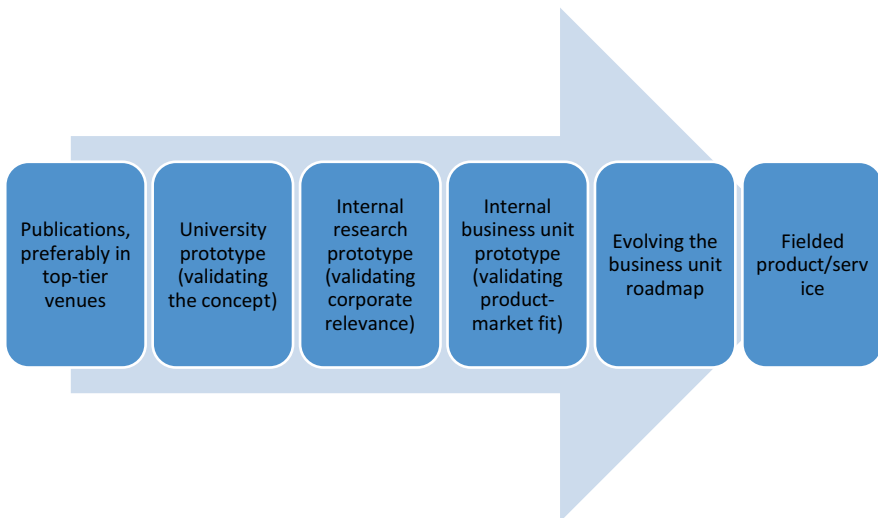


Fig. 12.1 Research translation indicators. Intermediate steps can be skipped in some cases

When working with entrepreneurial faculty, there are multiple innovation pathways available. Consider the example of Google and the research involving the PageRank algorithm (Page et al., 1998). Industrial collaborators in that research would have understood the algorithm well enough to incorporate the algorithm into their own products, an approach sometimes called “organic innovation,” but despite Stanford’s marketing efforts none licensed the patented algorithm. Ultimately the originating students leveraged the patented intellectual property and their know-how to create a company that could hypothetically have been acquired at some point by some larger company as an element of a contrasting “inorganic innovation” strategy. Neither pathway was taken; Google stood on its own and is now one of the most valuable companies in the world. In hindsight however, the value of a collaboration with that team could have been realized through either organic or inorganic innovation approaches. On the organic innovation pathway, an engineering team would have been the stakeholder. On the inorganic innovation pathway, an “corporate development” team (Corporate Finance Institute, 2024) would have been the stakeholder. On yet a third and intermediate pathway lies partnering, where the stakeholders would have been some combination of the two.

12.3.7 Broader Cross-Sector Partnering

Emerging technology is increasingly central to the evolution of many industries: biotechnology, energy, telecommunications, semiconductors, computing, and manufacturing, to name a few. Universities are key contributors to technology progress as well as talent creation. Despite the challenges of engaging with universities—timeframe alignment chief among these—successful industry leaders have a long tradition of engaging closely with universities and are generally motivated to continue doing so. Meanwhile, international rivalry and societal disruptions caused in part by the rapid pace of technological progress are causing governments to become more active in shaping and funding novel public–private partnerships. In the United States, the CHIPS and Science Act seeks to reinvigorate the semiconductor industry through a public–private partnership, the National Semiconductor Technology Center (NSTC), building from a long tradition of public–private partnerships in the semiconductor industry (Peña et al., 2021). The Act also authorized the National Science Foundation (NSF) to launch a new Technology Innovation Partnerships (TIP) directorate chartered with increasing the pace of research translation and the economic benefits of the government’s investment in basic research; public–private partnerships are a key tool of the TIP directorate.

An AET practitioner can leverage government partnering to increase the scale of available funding, which can help defray costs. At the same time, broader partnering

sometimes comes with complications that can hinder support within companies. A nonexhaustive list of considerations for multistakeholder collaboration includes:

- **Governance.** Multistakeholder decisions about the scope of solicitations, proposal review and selection, and post-award management can be challenging in a setting with representatives from multiple sectors.
- **Theory of competition.** Many public–private partnerships (PPPs) include companies that are explicitly competing with each other (Semiconductor Research Corporation, 2024). Competition and cooperation require careful management. Potential mitigations can include, for example, scoping the research focus in precompetitive time horizons, defraying the cost of common infrastructure, addressing ecosystem issues such as interoperability, or mitigating industry externalities (carbon footprint, public safety, etc.).
- **Technology strategy.** PPPs are often rooted in an explicit or implicit technology strategy, which can become a key topic of governance. A technology strategy for a PPP requires a deep understanding of the broader ecosystem and the role of other government and industry organizations.
- **Intellectual property.** When a PPP develops technology, the disposition of intellectual property can be a key factor in the success of the partnership.

As discussed in Section 12.5, PPPs are poised to become an even more important tool in evolving the national research enterprise.

12.4 Case Study: Inflection Points in the Semiconductor Industry

This case study illustrates how an Intel AET motivated the company to pursue an industry-scale research agenda through academic partnership, drawing on a number of ideas presented in this chapter.

In the late 2000s, the semiconductor and software industry faced a key inflection point. Driven up until that point by a combination of Moore’s “Law” and Dennard Scaling, microprocessor consumers saw remarkable and regularly-cadenced improvements in the capabilities of their personal computing based largely on increased clock speed. When that clock speed trend tapered for fundamental technical reasons, it became apparent that frequency scaling would need to be replaced by other foundations, in particular by increasing the number of processors rather than their clock speed. Unfortunately, this transition heralded a crisis in that software would need to be rewritten to take advantage of multicore scaling, whereas consumers had previously experienced the benefits of frequency scaling without any change to the underlying software. To address this industry-wide problem, Microsoft and Intel teamed up to launch a prominent, large-scale university research initiative called the “Universal Parallel Computing Research Centers” (UPCRC) (Green, 2008) and funded both the University of California at Berkeley and the University of Illinois, Champaign-Urbana, to do fundamental research in parallel programming.

The UPCRC initiative helped reorient the entire programming community at a crucial inflection point in semiconductor technology and was followed by sustained large-scale government-funded initiatives such as the NSF XPS program (National Science Foundation, 2012). Today, parallelism has been richly explored and relevant techniques are now incorporated into computer science curricula worldwide. Only mature industry leaders, such as Microsoft and Intel, would perceive the benefit of tackling industry-scale problems, and only mature industry leaders would be in a position to make sufficiently large-scale investment to “move the needle.”

The UPCRC case study is illustrative for UI research practitioners in multiple dimensions corporate evolution, governance, funding, and partnering. Clearly both sponsor companies recognized the need to help address an industry-wide need, and both were at a point in their own evolution where it was possible to make a needle-moving investment. However, the corporate mechanics were nontrivial. Up until the launch of UPCRC, a set of decentralized Intel “research councils” controlled independent university grant budgets on a topical basis, generally with small grants (Ramming, 2013). The incumbent research council system was ideal for addressing a variety of defined research areas through consensus mechanisms, but was not well positioned to recognize or respond to fundamental shifts in the industry structure. Budgets in that funding/governance regime had been largely static over time, without regard to industry or corporate context. In order to fund a large-scale UPCRC initiative, Intel Labs centralized a large fraction of its research council budgets and revamped the decision process to more easily follow from a *de novo* view of corporate research priorities. By centralizing some previously decentralized funding and introducing a new governance model, Intel positioned itself to tackle a strategic priority at scale.

Meanwhile, Intel’s manufacturing arm continued its quest to advance process technology with ever-lower feature sizes on a regular cadence. A key tool was and continues to be the Semiconductor Research Corporation (SRC), a long-standing PPP that funds pre-competitive university research and supports workforce development (Semiconductor Research Corporation, 2024). Today, dimensional scaling of process technology is reaching asymptotic limits (Kim, 2024). As a result, “fabless” semiconductor companies, such as Nvidia, are growing in value relative to the actual manufacturer (Intel, TSMC), because architectural differentiation offers enormous competitive advantage in important applications, such as artificial intelligence. Moreover, computing has become an increasingly important national security asset, prompting the bipartisan “CHIPS and Science” legislation intended to “strengthen American manufacturing, supply chains, and national security, and invest in research and development, science and technology, and the workforce of the future to keep the United States the leader in the industries of tomorrow” (White House, 2022). As a result, the National Semiconductor Technology Center (NSTC) is now being established (National Institute of Standards and Technology (NIST, *n.d.*)). For practitioners in the semiconductor and computing industries, the NSTC represents an important development and a forcing function to revisit UI strategies in ways that complement the work of the SRC.

Bottom line: The practitioner who can anticipate strategic inflection points in an industry and in a company will be better positioned to navigate change and shape

productive outcomes. UI engagement naturally provides the practitioner with the advanced insights needed to anticipate and address such change.

12.5 Takeaways and Future Strategies for Motivating UI Collaboration

The intent of this chapter has been to shine a spotlight on issues that can help the corporate practitioner navigate within a given corporate structure, in the context of a company's evolution, and in the context of broader industry issues. A quick recap:

- Align initiatives with the broader corporate philosophy and be alert to evolution in that philosophy.
- Select metrics that accurately reflect and track progress against goals that are meaningful in the context of an AET's function.
- Be alert to the implications of internal funding approaches and align initiatives accordingly. Alternately, shape those funding approaches to support the needed initiatives.
- Leverage governance structures to create buy-in and support for important initiatives.
- Structure collaboration agreements that harmonize timescales and goals for both the company and the university.
- Be versed in the many facets and approaches to “research translation,” given its prominence as a motivation for university collaboration.
- Be alert to broader opportunities for broader cross-section partnering.

Armed with these insights, the practitioner can motivate collaboration and play a meaningful role in shaping a company's opportunities.

Looking beyond current practice, the practitioner may also need to consider nation-scale and global context. The mid-2020s are seeing a confluence of disruptive trends in the US research enterprise that call for thoughtful response by UI collaboration practitioners. As documented in the NSF's biennial report on the state of U.S. Science and Engineering (National Science Board, [2024a](#)), US leadership in science and technology faces increasingly stiff competition (especially from China) while the US educational apparatus is failing to prepare a future scientific workforce that leverages the full potential of the US talent base with the proficiency needed to remain a STEM leader. More fundamentally, there are structural problems in the division of labor between government, industry, academia, and the philanthropic sector. Some data points (National Science Foundation, [2024b](#); National Science Board, [2024](#)) concerning international competition:

- China is the top overall producer of Science & Engineering (S&E) publications (crossover in 2017).
- China is growing rapidly in publication quality as assessed by its fraction of “top 1%” of highly cited publications.

- China is trending to surpass the United States in R&D expenditures.
- China has surpassed the United States in patenting activity.
- China has the greater knowledge and technology-intensive (KTI) manufacturing output.
- China is now awarding more S&E degrees than the United States.

Some data points concerning the US S&E workforce:

- Only 50% (approximately) of graduate and postdoctoral students/researchers are US citizens.
- Roughly 30% of master's and doctor degree students have temporary visas.
- Women and minorities continue to be under-represented in the science and engineering workforce.

Some data points concerning structural changes in the research enterprise:

- US R&D funding by industry has outpaced the growth in other sources.
- Although growing in absolute terms, the proportion of federal funding for basic research has dropped (since 2011) from 65% of the total to 40%, while industry has grown from 20% to 35%.

Recognizing these challenges, the National Academy of Sciences identified UI partnership as a key opportunity for progress in its 2024 “State of Science Address” (McNutt, 2024). Meanwhile, the National Science Board (Reed, 2024) provided congressional testimony calling for the mobilization “of all the country’s assets – government (at all levels), industry, academia, and nonprofits” through a more actionable National Technology Strategy. Dr. Reed’s testimony included these calls to action:

To stay competitive, we will need more coordination and partnership among government, academia, and industry to speed the translation of research, develop talent in areas of national need, seed STEM-based economic development nationwide, and inject critical technologies into our national defense ecosystem.

The radical changes in the R&D landscape call for a new approach that maintains the benefits of our distributed system while still positioning our nation for competitive success. Facilitating a truly national S&T strategy will require the ability to effectively and efficiently execute a whole-of-government effort, prioritize resources, and work hand-in-hand with our industry and academic partners, while being held accountable for progress toward agreed upon national goals.

A key enabler of radical change and progress has been the 2022 CHIPS and Science Act, which appropriated tens of billions in funding for the semiconductor industry, initiatives such as a National Semiconductor Technology Center, which is creating opportunities for UI collaboration to advance the state of the art. The act also authorized enormous sums for R&D and innovation including a new “Directorate for Technology, Innovation, and Partnerships” within the National Science Foundation. Finally, the act calls for the development of a 4-year comprehensive national science and technology strategy, under the guidance of the Office of Science and Technology Policy (White House Office of Science and Technology Policy, 2024). These all

represent meaningful opportunities for UI practitioners to formulate novel and meaningful collaborations (public–private partnerships in particular) using the principles developed in this chapter. With tools of engagement and visibility into the horizons of research, the industry practitioners of UI engagement can be leaders in helping corporations contribute and compete more effectively.

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Chris Ramming Over the course of his career, Ramming has worked extensively at the intersection of industry, academia, and government. At DARPA, Intel, and VMware he led programs to invest in a broad range of academic research collaborations across networking, cloud computing, security and privacy, artificial intelligence, and sustainability. His research investments spanned individual grants, large centers, multinational engagements, and joint programs with government and other companies. To realize meaningful research results in new products and services, much of Ramming's work has involved the translation and incubation of ideas emerging from those investments. He presently serves in an advisory capacity to organizations that leverage and facilitate collaborative R&D, including UIDP.org, DARPA's Information Science and Technology Study Group (ISAT), CRA.org/industry, Natcast.org, and the Confidential Computing Consortium. Ramming received a BA in computer science from Yale College and an MS in computer science from UNC Chapel Hill.